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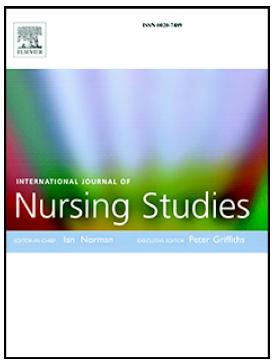

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Hospital Nurse Staffing Variation and Covid-19 Deaths: A Cross-sectional Study

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ABSTRACT

Background: During the Covid-19 pandemic, Covid-19 mortality varied depending on the hospital where patients were admitted, but it is unknown what aspects of hospitals were important for mitigating preventable deaths.

Objective: To determine whether hospital differences in pre-pandemic and during pandemic nursing resources—average patient-to-registered nurse (RN) staffing ratios, proportion of bachelors-qualified RNs, nurse work environments, Magnet recognition—explain differences in risk-adjusted Covid-19 mortality; and to estimate how many deaths may have been prevented if nurses were better resourced prior to and during the pandemic.

Methods: This is a cross-sectional study of 87,936 Medicare beneficiaries (65-99 years old) hospitalized with Covid-19 and discharged (or died) between April 1 and December 31, 2020, in 237 general acute care hospitals in New York and Illinois. Measures of hospital nursing resources (i.e. patient-to-RN staffing ratios, proportion of bachelors-qualified RNs, nurse work environments, Magnet recognition) in pre-pandemic period (December 2019 to February 2020) and during (April to June 2021) were used to predict in-hospital and 30-day mortality using adjusted logistic regression models.

Results: The mean age of patients was 78 years (8.6 SD); 51% were male **(n = 44,998)**. 23% of patients admitted to the hospital with Covid-19 died during the hospitalization **(n = 20,243);** 31.5% died within 30-days of admission **(n = 27,719)**. Patients admitted with Covid-19 to hospitals with better nursing resources pre-pandemic and during the pandemic were statistically significantly less likely to die. For example, each additional patient in the average nurses' workload pre-pandemic was associated with 20% higher odds of in-hospital mortality (OR 1.20, 95% CI [1.12- 1.28], p<0.001) and 15% higher odds of 30-day mortality (OR 1.15, 95% CI [1.09- 1.21], p<0.001). Hospitals with greater proportions of BSN-qualified RNs, better quality nurse work environments, and Magnet recognition offered similar protective benefits to patients during the pandemic. If all hospitals in the study had superior nursing resources prior

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to or during the pandemic, models estimate many thousands of deaths among patients hospitalized with Covid-19 **could have been avoided**.

Conclusions: Patients with Covid-19 admitted to hospitals with adequate numbers of RNs caring for patients, a workforce rich in BSN-qualified RNs, and high-quality nurse work environments **(both** prior to and during the Covid-19 pandemic**) were more likely to survive the hospitalization**. Bolstering these hospital nursing resources during ordinary times is necessary to ensure better patient outcomes and emergency-preparedness of hospitals for future public health emergencies.

Key words: nurse staffing, nurse work environment, mortality, Covid-19 pandemic, hospital

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What is already known about the topic:

- There is variation observed in mortality for patients admitted to hospital**s** with Covid-19 that is not fully accounted for by the clinical complexity of patients.
- A large body of research has linked **better** hospital nursing **resources** with better patient outcomes – but little is known about whether these hospital nursing factors might also explain the variation observed in hospital mortality for Covid-19 patients.

What this paper adds:

- Patients had significantly lower odds of dying if they attended hospitals with more favorable patient-to-RN ratios, higher proportions of bachelors-qualified RNs, better nurse work environments, and were Magnet recognized preceding the first U.S. wave of Covid-19 and during the pandemic.
- Ensuring hospitals have **appropriate** evidence-based RN resources during usual times **was** associated with better patient outcomes and reductions in deaths during **a** public health emergen**cy**.

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INTRODUCTION

Mortality rates among individuals hospitalized with Covid-19 varied across hospitals during the pandemic;(Asch, Islam et al. 2021, Asch, Sheils et al. 2021) however, there has been little explanation of this variation and what hospital features can be targeted to reduce preventable deaths during future public health emergencies. In this study, we investigated a possible explanation for why Covid-19 mortality varied across hospitals—chronic nurse understaffing and under-resourcing before and during the pandemic.(Lasater, Aiken et al. 2020, Aiken, Sloane et al. 2023)

During Covid-19 surges, the public witnessed nurses working to provide effective care despite untenable working conditions—widespread staff shortages and layoffs, rationed and reused PPE, and self-quarantine to avoid bringing Covid-19 home to families.(Times 2024) But even before the pandemic, nurses in many hospitals struggled with burnout, impossible workloads, poor management, and systemic breakdowns.(Lasater, Aiken et al. 2020, Aiken, Sloane et al. 2023) For example, data collected just prior to the Covid-19 pandemic documents chronic hospital nurse understaffing and wide variation in patient-to-registered nurse (RN) staffing, varying from 3.3 to 9.7 patients per RN on general medical-surgical units.(Lasater, Aiken et al. 2020) Nearly half of physicians and over 90% of nurses working in hospitals agree that improving nurse staffing would improve clinician burnout.(Aiken, Lasater et al. 2023)

Indeed, literature demonstrates that poorer hospital nursing resources explain poor clinician and patient outcomes. For example, nurse understaffing is predictive of higher patient mortality,(Aiken, Clarke et al. 2002, Aiken, Cimiotti et al. 2012, Aiken, Sloane et al. 2014, Lasater, Aiken et al. 2021, Lasater, Aiken et al. 2021, McHugh, Aiken et al. 2021) particularly among high-risk patients like those with sepsis.(Lasater, Sloane et al. 2020, Dierkes, Aiken et al. 2021, Lasater, McHugh et al. 2021, Cimiotti, Becker et al. 2022, Dierkes, Aiken et al. 2022) High proportions of bachelors of science in nursing (BSN)-qualified RNs,(Aiken, Clarke et al. 2003, Yakusheva, Lindrooth et al. 2014, Lasater, Sloane et al. 2021) quality nurse work environments,(Aiken, Clarke et al. 2008, Friese, Lake et al. 2008, Lake, Sanders et al. 2019) and Magnet recognition(Aiken, Smith et al. 1994, McHugh, Kelly et al. 2013, Friese, Xia et al. 2015) are

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associated with lower mortality as well as a host of other patient outcomes (e.g., complications, readmissions, length of stay, patient satisfaction, and safety events).(Kane, Shamliyan et al. 2007) Despite the evidence, attention to hospital nurse staffing, workforce composition, and work environments are generally absent from strategic emergency preparedness planning priorities and from evidence-based hospital quality standards more generally.

This **novel s**tudy examine**s** how variation in key features of hospital nursing impacted outcomes of patients hospitalized with Covid-19. Using data about hospital nursing resources collected in the weeks prior to the first wave of patients with Covid-19 entering U.S. hospitals, we evaluate**d** whether variation in hospitals' nursing was associated with **Covid** mortality. Using pre-pandemic measures of hospital nursing resources enables an evaluation of *preparedness* since it evaluates at a cross-section in time the baseline levels of staffing, work environments, and proportions of BSN-qualified RNs in a hospital during 'usual' times. Since hospitals likely modified staffing and nursing workforce configurations in response to the pandemic, we also evaluate**d** during-pandemic nursing resources on Covid-19 outcomes, in a sensitivity analysis. Finally, we estimate**d** how many in-hospital Covid-19 deaths might have been prevented if hospitals entered the pandemic with staffing levels, BSN-qualified workforces, and work environments that reflected recommendations of decades of research, Institute of Medicine reports,(Institute of Medicine 2000, Institute of Medicine 2004, Institute of Medicine 2011) and proposed federal and state policies.

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METHODS

Data. Unique primary survey data were collected via a survey of all RNs licensed in New York and Illinois just prior to the pandemic (December 16, 2019 through February 24, 2020) to obtain measures of staffing, nurse work environment, and BSN-qualified RNs in hospitals. These data were linked with Medicare Provider Analysis and Review (MedPAR) files of inpatient claims between April 1, 2020 and

December 31, 2020, and with data about hospital characteristics from the American Hospital Association (AHA) Annual Survey and the American Nurses Credentialing Center (ANCC) on Magnet recognition status.

We surveyed by email 100% of the actively licensed RNs in New York and Illinois. The survey was approved by the University of Pennsylvania institutional review board (Protocol # 834307). RNs received information about the purpose of the survey, that participation was voluntary, responses were anonymous, and completion of the survey represented informed consent. Non-respondents received repeated follow-up invitations at regular intervals. The average respondent completed the survey in less than 15 minutes. Nurses who indicated that they were employed in a hospital were asked to provide the name of their hospital, which enabled aggregation of nurse responses to create hospital-level measures of nursing resources. Using front-line nurses as informants of hospital nursing resources is an approach validated over decades of research(Aiken, Clarke et al. 2002) and shown to produce unbiased and representative estimates.(Lasater, Jarrín et al. 2019)

Patient Sample. Medicare beneficiaries admitted to a hospital in New York or Illinois during 2020 with a primary or secondary diagnosis for Covid-19 (ICD-10 U07.1) were included. **Medicare patients between the ages of 65-99 years were included to be consistent with studies of Medicare beneficiaries and since the majority of individuals hospitalized with Covid-19 were from this age group.**(Asch, Sheils et al. 2021) The sample included both traditional fee-for-service Medicare beneficiaries and those covered under Medicare Advantage. Patients were excluded if they were discharged from the hospital prior to April 1, 2020 since the use of ICD-10 U07.1 was only adopted for discharges occurring on and after that date. The final sample included 87,936 patients.

Hospital Sample. Adult non-federal acute care hospitals and critical access hospitals in New York and Illinois were included in the hospital sample if they had at least one admission for Covid-19 during the study period. The final hospital sample included 237 hospitals; with the average hospital represented by

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64 nurses working in direct patient care positions to report on the nursing resources in the hospital (range: 10-575 nurse informants per hospital).

Outcomes. In-hospital mortality was defined as a death that occurred during the hospitalization. 30-day mortality was defined as a death that occurred 30 days from admission, including both deaths that occurred during the hospitalizations and after discharge.

Predictors. The three hospital nursing resources that were derived from nurse survey informants were: patient-to-RN staffing ratios, quality of the nurse work environment, and hospital proportion of BSN-qualified RNs. Hospital-level patient-to-RN staffing ratios were derived from direct care RNs working on medical, surgical, and oncology units. Direct care RNs reported how many patients and how many RNs were on the unit during their last shift. Responses were aggregated by taking the average of the responses among nurses working in the same hospitals to create the hospital-level measure of patient-to-RN staffing ratios. This measure of staffing has advantages over alternative staffing data, since the measure we used only **included** RNs working in direct patient care whereas other staffing measures include nurses in non-direct care roles resulting in underestimates of direct care nurses' workload. The decision was made to evaluate patient-to-RN staffing ratios on medical-surgical units (as opposed to intensive care units (ICUs)) because fewer than 25% of patients hospitalized with Covid-19 were admitted to an ICU during our study period and the majority of patients who died during the hospitalization died outside of an ICU (**Appendix Table 1**).

The quality of the nurse work environment is measured using an abbreviated 5-item short-form of the National Quality Forum endorsed Practice Environment Scale of the Nursing Work Index.(Lake 2002) Nurse respondents were asked to rate on a 4-point Likert scale ranging from 'strongly agree' to 'strongly disagree' whether the following are present in their job: administration that listens and responds to employee concerns, a nurse manager who is a good manager and leader, a lot of team work between nurses and physicians, enough staff to get the work done, a clear philosophy of nursing that pervades the patient care environment. A hospital-level nurse work environment score, ranging from 1 (poorest

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environment) to 4 (best environment) was created by taking the average score among nurses in the same hospital.

For each hospital, the proportion of RNs with a bachelor of science in nursing (BSN) degree or higher was computed by aggregating the nurse respondent's reports of their highest level of education in nursing. Hospitals were defined as being Magnet-recognized if they were indicated as having an active Magnet status in 2019 on the ANCC website.

Patient Covariates. Patient covariates included age, sex, comorbidities, a dummy variable for whether the patient was admitted from a skilled nursing facility, discharge month, and a continuous variable of the number of days from March 1, 2020 to admission date. Consistent with prior research,(Asch, Islam et al. 2021, Asch, Sheils et al. 2021) we control**led** for the days from March 1 to hospital admission date to account for rapid clinical care improvements that occurred overtime. The Elixhauser Comorbidity Index for risk of in-hospital mortality, which is a composite score based on 38 individual comorbidity measures,(Moore, White et al. 2017) was used for patient risk adjustment. Indicator variables if present on admission (POA) for an encounter for palliative care or do not resuscitate (DNR) order were also included.(Cimiotti, Becker et al. 2022)

Additional patient risk adjustment was performed by using an external training dataset of patient data from Pennsylvania and Ohio to identify additional comorbidities that were predictive of in-hospital mortality among patients admitted with Covid-19. Those states were chosen for their geographic proximity and similar Covid-19 mortality rates to the study states—New York and Illinois. Using the training dataset, we applied the same patient inclusion/exclusion criteria as for the main study, and then identified 14 diagnoses codes that were present on admission and with frequencies greater than 10,000. Three of the 14 comorbidities overlapped with Elixhauser's and were excluded from further consideration. Step-wise regression excluded 1 additional comorbidity since it was not predictive of in-hospital mortality. The result was 10 diagnoses codes which were added to the regression modeling as indicator variables for patient risk adjustment.

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Hospital Covariates. Hospital size was categorized as small (≤100 beds), medium (101-250 beds), or large (>250 beds). Teaching status was defined by the proportion of medical residents/fellows to beds and was either non-teaching (no residents/fellows), minor teaching (<1:4 residents/fellows per bed) or major teaching (≥1:4 residents/fellows per bed). High technology hospitals were those with capabilities of performing open-heart surgery or major organ transplantation. **Hospital mean nurse years of experience was defined as the hospital-level average number of years nurse survey respondents said they have worked as a nurse.**(Zaranko, Sanford et al. 2023, Peutere, Pentti et al. 2024)

Statistical Analysis. Descriptive statistics **were** used to describe patient sample characteristics overall and among the subset of patients who experienced in-hospital and 30-day mortality; as well as to describe hospital characteristics. Logistic regression models, fully-adjusted for patient and hospital covariates, and accounting for clustering of patients within hospitals using Huber-White sandwich estimators,(Freedman 2006) provide**d** odds ratios for the independent effect of each nursing resource variable on each of the mortality outcomes. C-statistics **we**re reported for each model.

Post-hoc estimates enabled us to estimate how many fewer deaths would have occurred if all study hospitals met minimum requirements on each of the nursing resources. Specifically, we estimate**d** how many fewer Covid-19 deaths would have occurred if all study hospitals had a minimum ratio of 4 patients per RN on medical-surgical units, consistent with proposed staffing policies. We estimate**d** how many fewer Covid-19 deaths would have occurred if all study hospitals had a minimum of 80% BSN-qualified RNs, consistent with the recommendations made in the 2010 Institute of Medicine *Future of Nursing Report*.(Institute of Medicine 2011) Finally, we estimate**d** how many fewer Covid-19 deaths would have occurred if all study hospitals were in the top-quartile of the nurse work environment score; and if all hospitals were Magnet-recognized.(Kutney-Lee, Stimpfel et al. 2015, McHugh, Aiken et al. 2016) Statistical tests were 2-tailed and P values less than .05 were considered statistically significant. Data analyses were performed from May 11 to August 3, 2023, using Stata, release 17 (StataCorp LLC).

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Sensitivity Analysis

The main analysis of this study **evaluated** pre-pandemic hospital nursing resource measures associated with Covid-19 outcomes; we also evaluate**d** measures of patient-to-RN staffing ratios, quality of the nurse work environment, and hospital proportion of BSN-qualified RNs collected *during* the pandemic (April 13, 2021 through June 22, 2021**, Appendix Table 2**) via the same survey methodology previously described.

Ethical Approval

The study was approved by the University of Pennsylvania institutional review board (Protocol # 851908).

RESULTS

The patient sample included 87,936 Medicare beneficiaries admitted to one of the 237 New York or Illinois study hospitals with a diagnosis of Covid-19 and discharged between April 1 and December 31, 2020. Approximately half (51.2%**, n = 44,998**) of the patients were male, and the average age was 78 years old (SD 8.6 years).

Almost one-quarter (23%**, n = 20,243**) of patients hospitalized with Covid-19 died during the hospitalization (**Table 1**). Roughly one-third (31.5%**, n = 27,719**) of patients died within 30-days of admission including deaths after discharge. Both in-hospital and 30-day mortality were higher among older patients ($p<0.001$), males ($p<0.001$), and patients who were admitted to the hospital from a skilled nursing facility ($p<0.001$). Mortality was highest in April 2020 (36.1% in-hospital mortality**, n = 10,684**; 44.6% 30-day mortality**, n = 13,192**), with a nadir in July (9.3% in-hospital mortality**, n = 209;** 16.4% 30-day mortality**, n = 369**), followed by a steady rise through December 2020.

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On average, patients who died had a higher comorbidity burden, as represented by the **higher** Elixhauser Comorbidity Index score (**Table 2,** p<0.001). With respect to comorbidities and conditions present on admission, the most common co-occurring condition was for 'other viral pneumonia,' which was present in 76.2% of Covid-19 admissions **(n = 66,997)**. While only 4.5% of patients **(n = 3,915)** admitted to the hospital with Covid-19 had an associated ICD-10 code for an encounter for palliative care, 56.1% **(n = 2,196)** and 78.5% **(n = 3,074)** of these patients died in the hospital or 30-days from admission, respectively.

Across the 237 study hospitals, the average patient-to-RN staffing ratio on an adult medical-surgical unit was 5.6 patients per nurse (SD 1.0; interquartile range: 4.9 to 6.2; **Table 3)**. The average hospital had a nurse work environment score of 2.7 (on a scale of 1-4, with 4 being the most favorable score). The average hospital had an RN workforce comprised of 67.1% BSN-qualified nurses (SD 19.0%; interquartile range: 54.8% to 82.6%). Sixty-six (27.9%) of the 237 study hospitals were Magnet recognized. About half of hospitals (48.5%) had more than 250 beds, the greatest percentage of them were non-teaching hospitals (43%) and were mostly evenly distributed across technology status categories and states.

In fully adjusted models presented in **Table 4**, we used measures of each hospital's nursing resources *prior* to the pandemic to evaluate the outcomes of Covid-19 patients in those hospitals *during* the pandemic. Covid-19 patients who were admitted to hospitals in which nurses had higher workloads, poorer nurse work environments, and lower proportions of BSN-qualified nursing staff *prior* to the pandemic, were statistically significantly more likely to die during their hospitalization for Covid-19 or within 30-days of admission.

Specifically, each additional patient in the average nurses' workload prior to the pandemic was associated with 20% higher odds of in-hospital mortality and 15% higher odds of 30-day mortality for patients during the pandemic (**Table 4**). Patients admitted to hospitals that were 1 SD above the average with respect to the nurse work environment prior to the pandemic were 15% and 12% less likely to experience

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in-hospital and 30-day mortality, respectively. Similarly, higher proportions of BSN-qualified nurses in hospitals prior to the pandemic was associated with statistically significantly lower odds of mortality for Covid-19 patients in those hospitals during the pandemic. Magnet hospitals had a protective effect on Covid-19 patient outcomes during the pandemic, with patient mortality odds 23**%**-25% lower in Magnet hospitals compared to non-Magnet hospitals. C-statistics for the regression models were 0.8, suggesting a strong model fit.

Since previous studies have shown that these hospital nursing resources are especially beneficial for patients with high risk for mortality,(Silber, Rosenbaum et al. 2016, Lasater, McHugh et al. 2021, Lasater, McHugh et al. 2021) we tested interactions of the nursing resources on mortality outcomes by patients' Elixhauser Comorbidity Index score. We found that Covid-19 patients with greater comorbidity burden had lower odds of in-hospital and 30-day mortality in hospitals with higher proportions of BSN-qualified nurses prior to the pandemic (OR 0.97, $p<0.001$ in-hospital mortality; OR 0.97, $p<0.001$ 30-day mortality); suggesting that more clinically complex patients experience an outcomes advantage of being cared for in hospitals with greater proportions of highly-educated nurses. None of the other interaction terms achieved statistical significance.

Using the risk-adjusted coefficients from our regression models, we estimated the number of preventable in-hospital deaths associated with Covid-19 if all the study hospitals achieved a minimum threshold of nursing resources prior to the pandemic (**Table 5**). For example, if all the study hospitals staffed not more than 4 patients per RN, our models estimate 3,**460** in-hospital Covid-19 deaths could have been prevented during the 9-month study period in the 2 study states – this translates into a 17% reduction in mortality. **Six hundred and thirty-one** deaths could have been prevented if all hospitals employed a workforce comprised of a minimum of 80% BSN-qualified RNs. If hospitals had nurse work environments in the top 75th percentile or Magnet recognition, we estimate**d** 1,7**01** and **1,984** preventable deaths, respectively.

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Results of Sensitivity Analysis

Using hospital nursing resources measures collected during the pandemic, the results of the fully adjusted models demonstrate that each additional patient in a nurses' workload was associated with **10**% higher odds of in-hospital mortality and **7**% higher odds of 30-day mortality (**Appendix Table 3**). A 1 standard deviation improvement in the nurse work environment was associated with reductions of **13**% and 10% for in-hospital and 30-day mortality, respectively. And a 10% increase in the proportion of BSN-qualified RNs was associated with reductions of **7**% and **6**% for in-hospital and 30-day mortality, respectively. Comparisons of odds ratios between pre-pandemic measures of hospital nursing resources (**Table 4**) and during-pandemic measures (**Appendix Table 3**) demonstrate effect sizes of similar magnitudes and significance.

DISCUSSION

This cross-sectional study of 237 hospitals and 87,936 older adult patients admitted with Covid-19 finds that wide variation in hospital nursing resources prior to the Covid-19 pandemic is associated with many thousands of preventable Covid-19 deaths during the pandemic.

Hospital nursing resources are often overlooked in conversations about lessons from the U.S. Covid-19 response(Covid Crisis Group 2023) and hospitals have not yet been able to recover from the effects of Covid as they continue to experience difficulties recruiting and retaining nurses after the official end of the pandemic.(Aiken 2023, Aiken, Sloane et al. 2023) For example, previous research demonstrates that hospitals with poor staffing and work environments prior to the pandemic had higher nurse burnout, job dissatisfaction, intent to leave employment among nurses one year into the pandemic; as well as poorer nurse-reported quality of care measure like poor grades on patient safety and infection prevention.(Aiken, Sloane et al. 2023)

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To our knowledge, this is the only study to evaluate the relationship between pre-pandemic modifiable aspects of hospital preparedness on patient outcomes during the Covid-19 emergency. In a different study undertaken during the pandemic, researchers found that both physicians and nurses ranked improving nurse staffing as a highly important intervention to reduce their experienced burnout.(Aiken, Lasater et al. 2023) Others have studied hospital nurse resources *during* the pandemic and found serious misalignments between the resources nurses were provided and what nurses said would have been most helpful to them—suggesting that even during the pandemic the U.S. response to supporting front-line RNs was lacking.(Cho, Sagherian et al. 2023) Specifically, the study by Cho and colleagues(Cho, Sagherian et al. 2023) found that the most important resources that nurses said they needed in order to provide safe and quality patient care during and after the pandemic were flexible work scheduling and adequate staffing; but more than two-thirds of nurses reported these resources were not provided to them by their hospitals. Greater national attention to organizational plans for securing nursing workforce resources is also echoed in qualitative studies of front-line nurses during the Covid-19 pandemic.(Norful, Tucker et al. 2023)

Limitations

Though the cross-sectional design of the study limits causal inferences about the relationship between hospital nursing resources and Covid-19 mortality outcomes, the use of pre-pandemic hospital nursing preparedness measures and pandemic outcomes is useful for informing future emergency preparedness efforts. We have studied almost all hospitals in two large U.S. states rather than sampling hospitals, which is a strength although these states and their patients might not be representative of all parts of the U.S. Estimated preventable deaths are conservative since study data were limited to Medicare patients **and future research should consider evaluating the outcomes of patients younger than 65 years old**.

Conclusions

Many deaths among patients hospitalized with Covid-19 could have been prevented if hospitals entered the pandemic with adequate numbers of RNs providing direct care, a workforce rich in BSN-qualified

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RNs, and high-quality nurse work environments. These modifiable aspects of hospitals' nursing resources are linked to more favorable patient outcomes during ordinary times and also represent important emergency-preparedness **dashboard metrics** that should **required in emergency preparedness plans.**

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Conflicts of Interests

None.

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Table 1. Mortality Outcomes and Characteristics of 87,936 Patients Hospitalized with Covid-19 between April and December 2020

	Overall N = 87,936		In Hospital Mortality N = 20,243		30 Day Mortality N = 27,719	
	N	**%**	**N**	**%**	**N**	**%**
Patient Outcomes						
In-hospital mortality	20,243	23.0	--	--	--	--
30-day mortality	27,719	31.5	--	--	--	--
Patient Characteristics						
Age*						
65-69	17,843	20.3	2,985	16.7	3,789	21.2
70-74	17,144	19.5	3,404	19.9	4,217	24.6
75-79	16,294	18.5	3,731	22.9	4,770	29.3
80-84	14,642	16.7	3,668	25.1	4,968	33.9
85-89	11,679	13.3	3,252	27.8	4,849	41.5
90-94	7,603	8.7	2,333	30.7	3,670	48.3
95-99	2,731	3.1	870	31.9	1,456	53.3
Sex*						
Male	44,998	51.2	11,366	25.3	15,222	33.8
Female	42,938	48.8	8,877	20.7	12,497	29.1
Admitted from skilled nursing facility*						
Yes	9,335	10.6	2,779	29.8	4,297	46.0
No	78,601	89.4	17,464	22.2	23,422	29.8
Discharge Month (2020)						
April	29,608	33.7	10,684	36.1	13,192	44.6
May	12,312	14.0	2,703	22.0	3,794	30.8
June	4,619	5.3	631	13.7	965	20.9
July	2,249	2.6	209	9.3	369	16.4
August	2,132	2.4	254	11.9	401	18.8
September	2,301	2.6	292	12.7	472	20.5
October	4,359	5.0	537	12.3	911	20.9
November	11,563	13.2	1,732	15.0	2,822	24.4
December	18,793	21.4	3,201	17.0	4,793	25.5
	Mean	**SD**	**Mean**	**SD**	**Mean**	**SD**
Days from 3/1/20 to Admission	143.9	105.8	108.7	100.8	119.8	103.7

Notes. *Mortality cases were significantly different from survivors at $p<0.001$

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Table 2. Patient Comorbidity Index and Conditions Present on Admission

	Overall N = 87,936		In Hospital Mortality N = 20,243		30 Day Mortality N = 27,719	
	Mean	**SD**	**Mean**	**SD**	**Mean**	**SD**
Elixhauser Comorbidity Index for risk of in-hospital mortality*	11.4	16.1	15.1	17.0	15.8	17.1
Conditions & Comorbidities, Present on Admission	**N**	**%**	**N**	**%**	**N**	**%**
Other viral pneumonia	66,997	76.2	17,894	26.7	23,454	35.0
Hyperlipidemia, unspecified	41,237	46.9	8,792	21.3	12,130	29.4
Acute respiratory failure with hypoxia	40,193	45.7	12,641	31.5	16,297	40.6
Acute kidney failure, unspecified	31,187	35.5	10,087	32.3	13,197	42.3
Atherosclerotic heart disease of native coronary artery without angina pectoris	23,618	26.9	5,810	24.6	8,006	33.9
Other specified sepsis	18,319	20.8	7,792	42.5	9,414	51.4
Do not resuscitate	17,458	19.9	6,766	38.8	9,840	56.4
Gastro-esophageal reflux disease without esophagitis	16,563	18.8	3,110	18.8	4,538	27.4
Dehydration	14,587	16.6	3,597	24.7	5,293	36.3
Acidosis	13,844	15.7	6,140	44.4	7,391	53.4
Hypokalemia	13,109	14.9	2,461	18.8	3,725	28.4
Encounter for palliative care	3,915	4.5	2,196	56.1	3,074	78.5

Notes. *Mortality cases were significantly different from survivors at $p<0.001$

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Table 3. Hospital Nursing Resources and Characteristics in 237 Study Hospitals

Nursing Resources	**Mean**	**SD**	**Interquartile Range**
Patient-to-RN staffing ratio	5.6	1.0	**4.9 - 6.2**
Nurse work environment score	2.7	0.3	**2.5 - 2.9**
Proportion of RNs with a BSN or higher	67.1%	19.0%	**54.8% - 82.6%**
	N	**%**	
Magnet-recognized	66	27.9	--
Hospital Characteristics	**N**	**%**	
Size			
Small	35	14.8	
Medium	87	36.7	
Large	115	48.5	
Teaching Status			
None	102	43.0	
Minor	80	33.8	
Major	55	23.2	
High Technology Capabilities			
Not-high	138	58.2	
High	99	41.8	
State			
New York	126	53.2	
Illinois	111	46.8	

Note. RN (registered nurse). BSN (bachelors of science in nursing). Staffing is derived from only 236 hospitals because 1 of the study hospitals did not have responses from direct care staff nurses working on a medical, surgical, or oncology unit to report reliable estimates of staffing.

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Table 4. Estimated Risk-Adjusted Odds of In-Hospital and 30-Day Mortality for Covid-19 Patients based on Hospital Nursing Resources Just Prior to the Pandemic

	In-hospital Mortality	**30-day Mortality**
	Odds Ratio [95% CI]	**Odds Ratio** [95% CI]
Each additional patient-per-RN (i.e. staffing ratio)	**1.20** [1.12 - 1.28] P <0.001	**1.15** [1.09 - 1.21] P <0.001
1 standard deviation improvement in nurse work environment	**0.85** [0.78 - 0.93] P <0.001	**0.88** [0.82 - 0.94] P = 0.001
10% increase in proportion of RNs with a BSN or higher	**0.92** [0.87 - 0.97] P = 0.003	**0.95** [0.91 - 0.99] P = 0.016
Magnet (vs. non-Magnet) hospital	**0.75** [0.64 - 0.88] P <0.001	**0.77** [0.68 - 0.87] P <0.001
c-statistic	**0.801**	**0.797**

Note. Models are fully-adjusted for patient covariates (age, sex, admission from skilled nursing facility, indicator for DNR order present on admission, indicator for palliative care encounter present on admission, Elixhauser Comorbidity Index, 10 indicator variables of additional comorbid conditions present on admission, discharge month, and number of days from admission date since 3/1/2020); and hospital covariates (size, teaching status, technology capabilities, **mean nurse years of experience**).

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Table 5. Estimated Preventable Covid-19 Deaths if Hospitals had Better Nursing Resources Prior to the Pandemic

	In-Hospital Mortality
Observed number of patients experiencing outcomes	**20,243**
Expected number of patient deaths if all hospitals staffed not more than 4 patients per RN	16,783
Difference between observed and expected	*3,460 preventable deaths*
Expected number of patient deaths if all hospitals had a minimum of 80% BSN-qualified RNs	19,612
Difference between observed and expected	*631 preventable deaths*
Expected number of patient deaths if all hospitals were in the top 75th percentile of work environment	18,542
Difference between observed and expected	*1,701 preventable deaths*
Expected number of patient deaths if all hospitals were Magnet	18,259
Difference between observed and expected	*1,984 preventable deaths*

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Declaration of interests

☐ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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